Lecture 4: Limits III

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Quiz Time!

Let's get into formation for quiz time.

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1. Recognizing when functions were *continuous* so that we could reduce the computation of limits to simpler evaluations
2. Developing limit properties which allowed us to build up the computation of complicated limits via reduction to simpler ones

We also developed the notion of the *one sided limits* - restricted kinds of limits where we need only approach from a singular side.

Today's Goals

We'll apply both our new limit computation skills as well as our knowledge of one-sided limits to work with *limits at or involving infinity*.
We will also uncover our last tool to help us compute limits: *the Squeeze Theorem*.

Limits at/involving Infinity

Prelude I

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And we wanted to compute $\lim_{x \to 1} g(f(x)) = \lim_{x \to 1} g((x-1)^2) = \lim_{u \to 0^+} g(u) = 3$

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Which allowed us to do a “substitution” that made the limit easier to compute.
This will be our strategy for computing limits at $\pm\infty$! Since $\pm\infty$ is not an actual number we can compute with in a consistent/coherent way, we’ll try and replace $x \to \pm\infty$ with a limit as *some other quantity* approaches *some actual number*.

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For a function $f(x)$, we say that $\lim_{x \to \infty} f(x) = L$ if we can make $f(x) \approx L$ (up to some error $\epsilon > 0$) just by forcing $x > N$ for some real number N.

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Do we want to prove things in this class like we would need to if we actually used this definition? NO!

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Similarly $x \to -\infty$ is *exactly equivalent* to $\frac{1}{x} \to 0^-$!

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Where we unfortunately encounter a problem since $\frac{1}{u^3}$ has an *asymptote* at $u = 0$ and so cannot be replaced by a function continuous at 0. What do we do?

Limits at Infinity V: Example 2 and Divergence to Infinity

Let's look at the graph!

Limits Diverging to Infinity

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For a function $f(x)$ and a point a, we say that $\lim_{x \to a} f(x) = +\infty$ (read as the limit of $f(x)$ as x approaches a diverges to positive infinity)

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Limits at Infinity V: Example 2 and Divergence to Infinity

Let's look at the graph!

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Then we must have that $\lim_{x \to a} f(x)$ exists and is equal to the common value of $\lim_{x \to a} g(x) = \lim_{x \to a} h(x)$.

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4. Inversion for Limits at infinity: $x \to \infty \iff \frac{1}{x} \to 0^+$, $x \to -\infty \iff \frac{1}{x} \to 0^-$
5. Squeeze Theorem: maybe you have a limit which involves a term that "shouldn't matter" intuitively

Tools for Algebraically Computing Limits

Let's recap our tools:

1. Evaluation for continuous functions: if $f(x)$ is continuous at $x = a$ then

$$f(a) \quad = \quad \lim_{x \to a^+} f(x) = \lim_{x \to a} f(x) = \lim_{x \to a^-} f(x) \tag{23}$$

2. Reduction to continuous functions: maybe the point at which you are taking the limit towards isn't in the domain but you can replace it with one and use tool #1
3. Limit Properties: You can rewrite your limit as a sum, product, quotient, or composite of other limits and reduce it to computing those
4. Inversion for Limits at infinity: $x \to \infty \iff \frac{1}{x} \to 0^+$, $x \to -\infty \iff \frac{1}{x} \to 0^-$
5. Squeeze Theorem: maybe you have a limit which involves a term that "shouldn't matter" intuitively and you can use the range of that term to put bounds that squeeze the value of the limit.